Inverse trigo Functions

One-one + onto
$\underbrace{\text{One-one + onto}}$

Bijective

$\Rightarrow$ Invertible

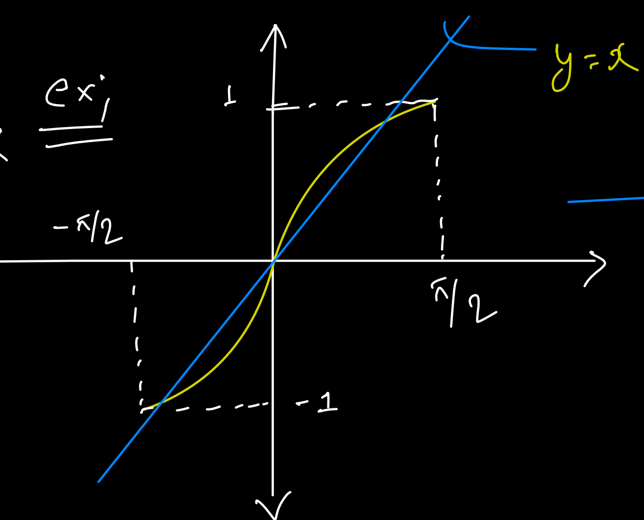
$T^{-1}(x) \longrightarrow$ unique angle

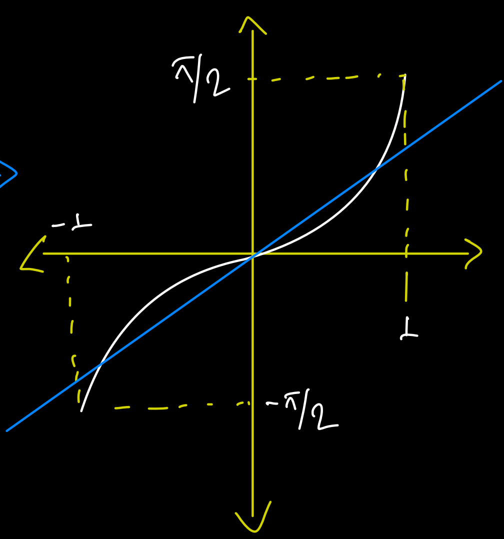
$$\boxed{T^{-1}(\underbrace{\ldots\ldots}_{\text{value}})} \longrightarrow \underline{\text{angle}}$$

** $T^{-1}(x)$ & $T(x)$ is symmetrical about $y = x$.

mirror image

ex:





Domains, Ranges & imp points

**

$\sin^{-1}(x) \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

$\tan^{-1}(x) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

$\text{cosec}^{-1}(x) \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$

I & IV

(+ve) (-ve)

$\cos^{-1}(x) \in [0, \pi]$

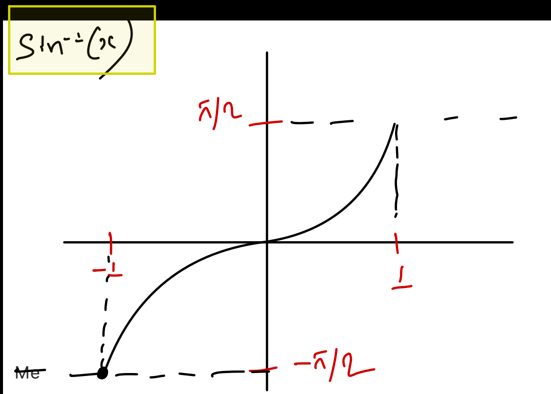
$\cot^{-1}(x) \in (0, \pi)$

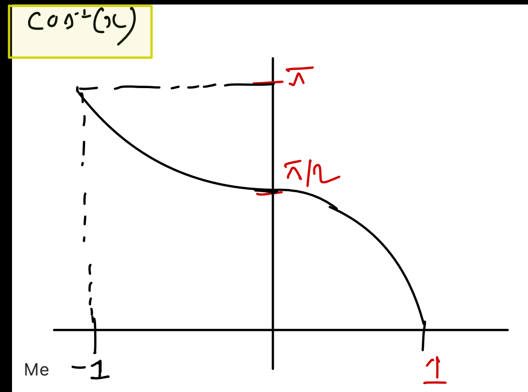
$\sec^{-1}(x) \in [0, \pi] - \{0\}$

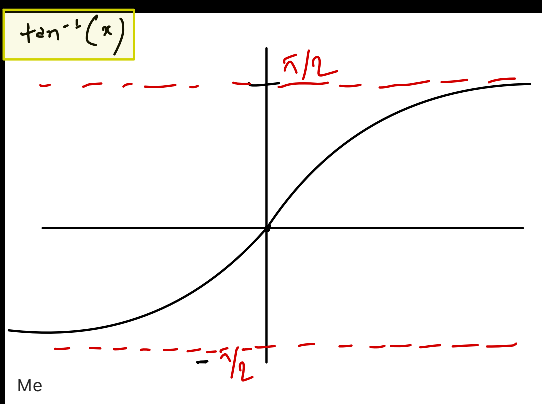
I & II

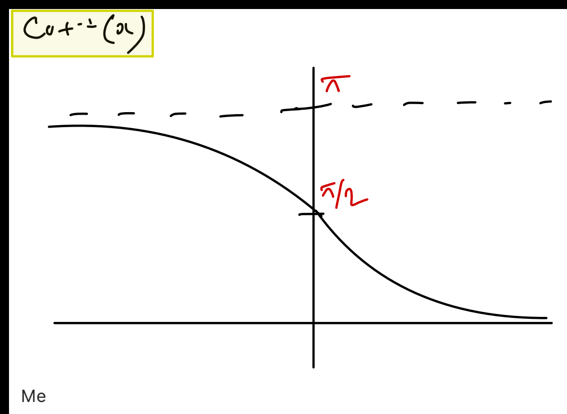
(+ve) (-ve)

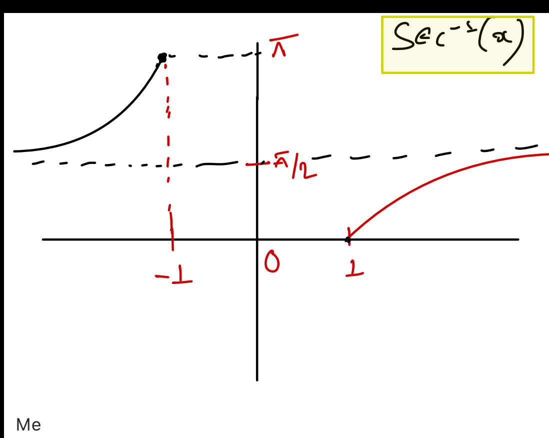
** Any values not lies in III - quadrant.

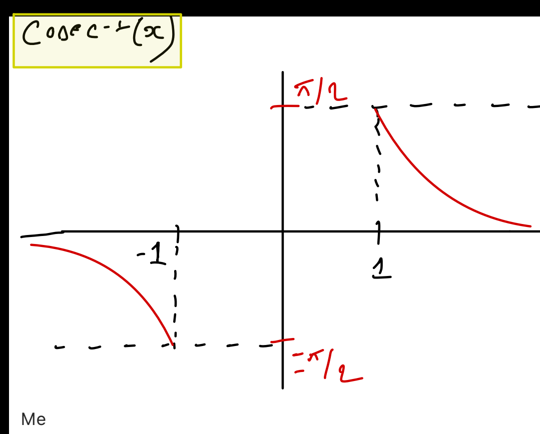

Sin⁻¹(x)
π/2
-1
1
-π/2
~~Me~~


Cos⁻¹(x)
π
π/2
-1
1
Me


tan⁻¹(x)
π/2
-π/2
Me


Cot⁻¹(x)
π
π/2
Me


Sec⁻¹(x)
π
π/2
-1
0
1
Me


Cosec⁻¹(x)
π/2
-1
1
-π/2
Me

properties

P1.)

$$F^{-1}(-x)$$

$\cos^{-1}(-x)$

$= \boxed{\pi - \cos^{-1}(x)}$

$$\boxed{-F^{-1}(x)}$$

$\sin^{-1}(x)$, $\operatorname{cosec}^{-1}(x)$

$\tan^{-1}(x)$

$$\boxed{\pi - F^{-1}(x)}$$

$\cos^{-1}(x)$, $\sec^{-1}(x)$

$\cot^{-1}(x)$.

P.2)

$$\boxed{F(F^{-1}(x))}$$

easy-peasy

$$y = \sin(\sin^{-1}(x)) = x$$

$$\cos(\cos^{-1}(x)) = x$$

$x \in [-1, 1]$

$$y = \tan(\tan^{-1}(x)) = x$$

$\forall\, x \in$ Domain

P.3)

$$\boxed{F^{-1}(F(x))}$$

<u>Basic;</u>

Let ex;

$$\sin^{-1}(\sin(x)) \neq \frac{7\pi}{4}$$

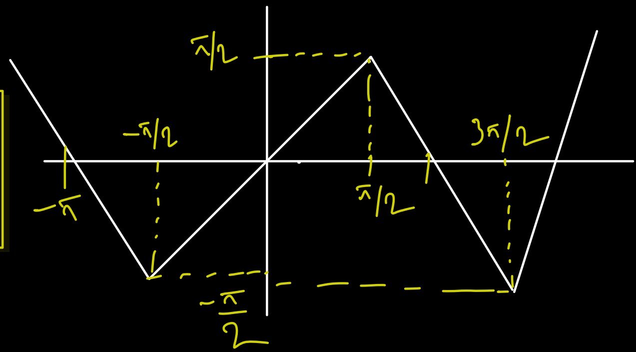
$$x = \frac{7\pi}{4}$$

$\Rightarrow$ $\sin^{-1}(\ldots)$ can only take values from $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

(i) $y = \sin^{-1}(\sin(x))$

$x \in R$

$$\boxed{y = \begin{cases} x & x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \\ \pi - x & x \in \left[\frac{\pi}{2}, \frac{3\pi}{2}\right] \end{cases}}$$

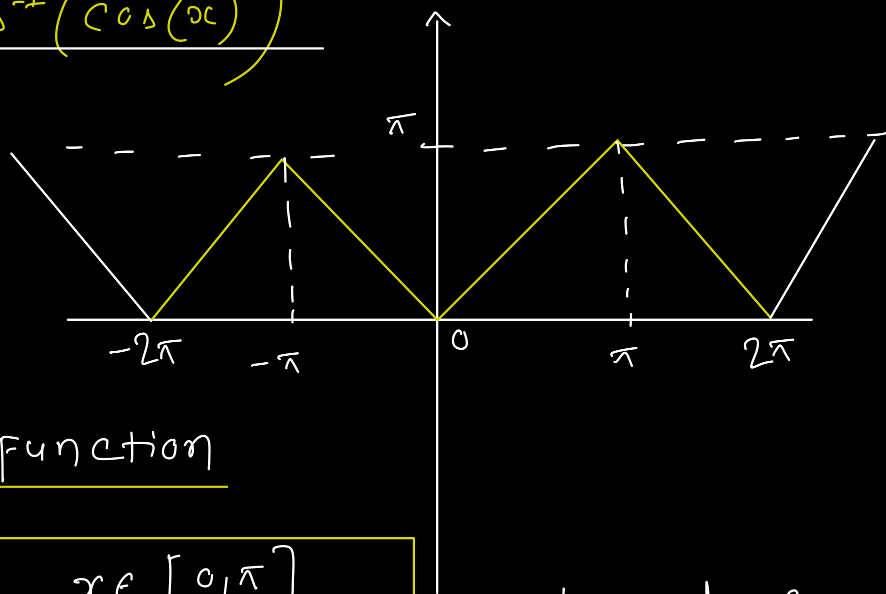


Continuous

$x \in R$

period $= 2\pi$

(ii) $y = \cos^{-1}(\cos(x))$



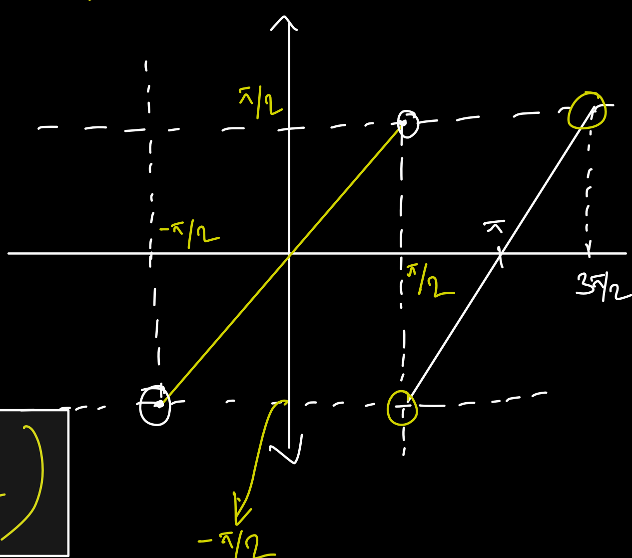
$x \in R$

Even function

$$\boxed{y = \begin{cases} x & x \in [0, \pi] \\ -x & x \in [-\pi, 0] \end{cases}}$$

period = 2π

(iii) $y = \tan^{-1}(\tan(x))$



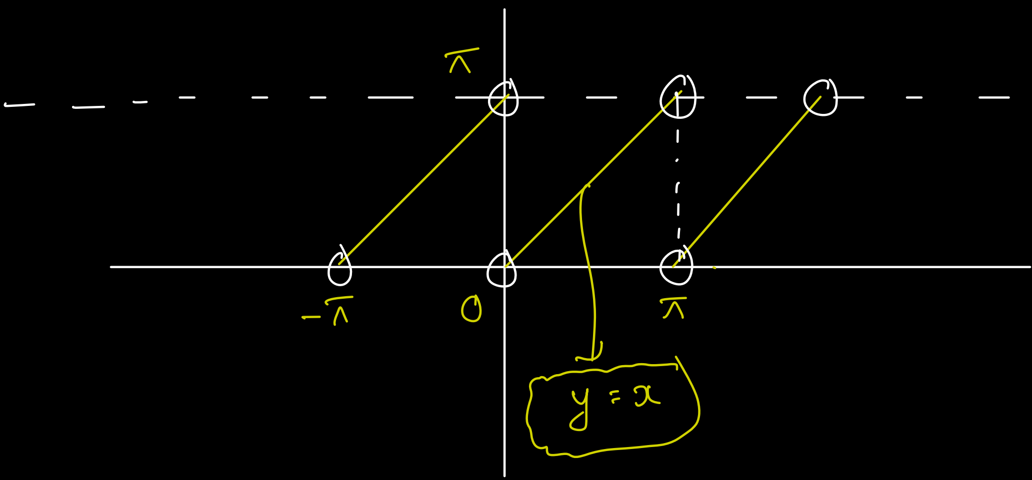
$x \in R - \{(2n+1)\frac{\pi}{2}\}$

odd function

period = π

$y = x \quad x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

(iv) $y = \cot^{-1}(\cot(x))$



py.) $$\boxed{\sin^{-1}(x) + \cos^{-1}(x) = \frac{\pi}{2} \qquad x \in [-1, 1]}$$

$$\boxed{\tan^{-1}(x) + \cot^{-1}(x) = \frac{\pi}{2} \qquad x \in R}$$

$$\boxed{\operatorname{cosec}^{-1}(x) + \sec^{-1}(x) = \frac{\pi}{2} \qquad |x| \geq 1}$$

$$\operatorname{cosec}^{-1}(x) = \sin^{-1}\left(\frac{1}{x}\right) \qquad |x| \geq 1,\ x \neq 0$$

$$\sin^{-1}(x) = \operatorname{cosec}^{-1}\left(\frac{1}{x}\right) \qquad |x| \leq 1,\ x \neq 0$$

$$\sec^{-1}(x) = \cos^{-1}\left(\frac{1}{x}\right) \qquad |x| \geq 1,\ x \neq 0$$

$$\cos^{-1}(x) = \sec^{-1}\left(\frac{1}{x}\right) \qquad |x| \leq 1,\ x \neq 0$$

$$\cot^{-1}(x) = \tan^{-1}\left(\frac{1}{x}\right) \qquad x > 0$$

$$= \pi + \tan^{-1}\left(\frac{1}{x}\right) \qquad x < 0$$

$$\Rightarrow \cot^{-1}(x) = p \in [0, \pi]$$

$$x = \cot(p)$$

$$\tan(p) = \frac{1}{x}$$

$$\tan^{-1}(\tan(p)) = \tan^{-1}\left(\frac{1}{x}\right)$$

$$p \qquad \left(\frac{\pi}{2}, \pi\right) \qquad p - \pi$$

Sum & diff. of ITF

(i) $\tan^{-1}(x) + \tan^{-1}(y)$

$x, y > 0$, $xy < 1$: $\boxed{\tan^{-1}\left(\dfrac{x+y}{1-xy}\right)}$

$x, y > 0$, $xy > 1$: $\boxed{\pi + \tan^{-1}\left(\dfrac{x+y}{1-xy}\right)}$

$\Rightarrow$ if $x, y < 0$

(ii) $\tan^{-1}(x) - \tan^{-1}(y)$

$\boxed{\tan^{-1}\left(\dfrac{x-y}{1+xy}\right)}$ $\quad xy > -1$

** Dusra ITF dikhe change in $\tan^{-1}(x)$ and enjoy.

Converting one ITF to another

ex; $\theta = \cot^{-1}(x)$

$x > 0$

$$\theta = \cos^{-1}\left(\frac{x}{\sqrt{x^2+1}}\right)$$

$x < 0$

$$\theta = \pi - \cot^{-1}(-x)$$

$$\Rightarrow \pi - \cos^{-1}\left(\frac{-x}{\sqrt{x^2+1}}\right)$$

$\boxed{\text{I.T.F equations}}$

(i) Apply same trigo functions.

(ii) verify your sol^n/values.

ex; $\underbrace{\tan^{-1}(\ldots)}_{A} - \underbrace{\tan^{-1}(\ldots)}_{B} = \underbrace{(\ldots)}_{C}$

tan lagao both side.

(iii) Domain & Range

$$\boxed{F(x) = \sin^{-1}(x) + \cos^{-1}(x) + \tan^{-1}(x) + \cot^{-1}(x) + \sec^{-1}(x) + \operatorname{cosec}^{-1}(x)}$$

$$\Rightarrow \left.\begin{array}{l} \sin^{-1}(x) \\ \cos^{-1}(x) \end{array}\right] \quad x \in [-1, 1]$$

$$\left.\begin{array}{l} \sec^{-1}(x) \\ \operatorname{cosec}^{-1}(x) \end{array}\right] \quad |x| \geq 1$$

$D_{F(x)} \in \{-1, 1\}$

Check.

ITF Series (phd Series).

Some Substitutions

$\sqrt{1+x^2}$

$x = \tan(\theta)$ or $\cot(\theta)$

$\sqrt{x^2-1}$

$x = \sec(\theta)$ or $\operatorname{cosec}(\theta)$

$\sqrt{1-x^2}$

$x = \sin(\theta)$ or $\cos(\theta)$

* Don't remember but main aim; remove root

* use trigo functions

$$\boxed{\sqrt{x^2} = |x|}$$

ex: Reduce $\tan^{-1}\left(\dfrac{\sqrt{1+x^2}-1}{x}\right)$

$x \in R - \{0\}$

$\Rightarrow$

$x = \tan(\theta)$

$\underbrace{\theta = \tan^{-1}(x)}_{\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)}$

$\tan^{-1}\left(\dfrac{\frac{1}{|\cos(\theta)|} - 1}{\frac{\sin(\theta)}{\cos(\theta)}}\right)$

$\dfrac{1-c(\theta)}{s(\theta)} \Rightarrow \underbrace{t\left(\frac{\theta}{2}\right)}$

$\tan^{-1}\left(\tan\left(\frac{\theta}{2}\right)\right)$

$\frac{1}{2}\tan^{-1}(x) = 1$

$\tan^{-1}(x) = 2$ ✓

$\underbrace{}_{x}$

Important Graphs

$x = \tan(\theta)$

$\sin^{-1}(\sin(\underbrace{2\theta}))$

$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

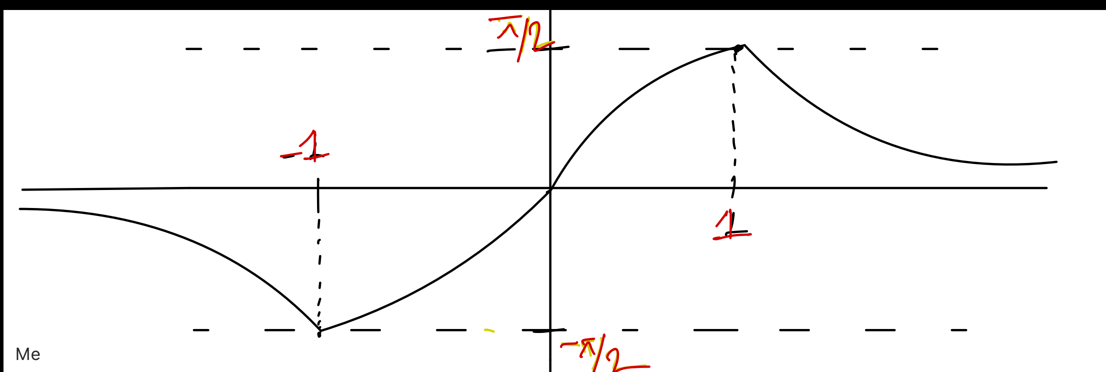
2θ

(i) $y = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$

$2\tan^{-1}(x)$ — $|x| \le 1$

$\pi - 2\tan^{-1}(x)$ — $x > 1$

$-\pi - 2\tan^{-1}(x)$ — $x < -1$



ex:

$$y = \sin^{-1}\left(\frac{8x}{1+16x^2}\right)$$

(a) $f(x)$ is $\uparrow$ in $x \in$ ____

(b) No. of Sol^n of $y = e, \frac{\pi}{8}, 2.2$ ____

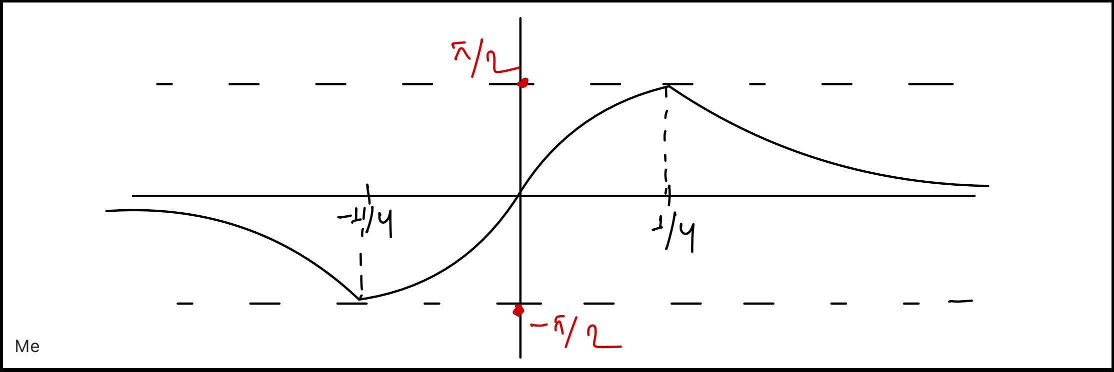
(c) point of maximum / minimum

(d) Range in interval

Approach

$$\boxed{x \to \frac{x}{4}}$$

$$y = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$$



(ii) $y = \cos^{-1}\left(\dfrac{1-x^2}{1+x^2}\right)$

$\boxed{\begin{matrix} 2\tan^{-1}(x) \\ x \geq 0 \end{matrix}}$

$-2\tan^{-1}(x)$

$x < 0$

even function